UNIVERSITÀ DEGLI STUDI FIRENZE

Can’t Get No Satisfaction

3-SAT Experimental Analysis

Statistical Physics

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Goals:

- **Satisfiability threshold** in the 3-SAT problem.
- **Ratio** between clauses and variables.
- **Efficiency** of search algorithms against industrial solvers.

The Boolean Satisfiability Problem (SAT):

Definition: a Boolean formula is **satisfiable** if there exists an assignment of values that makes the formula true.

Example:

$$(p \vee \neg q) \wedge (q \vee r) \wedge (\neg r \vee \neg p)$$

Brute Force: using truth tables requires 2^n steps.
For large n, the search becomes *"exhausting rather than exhaustive."*

p	q	r	form
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

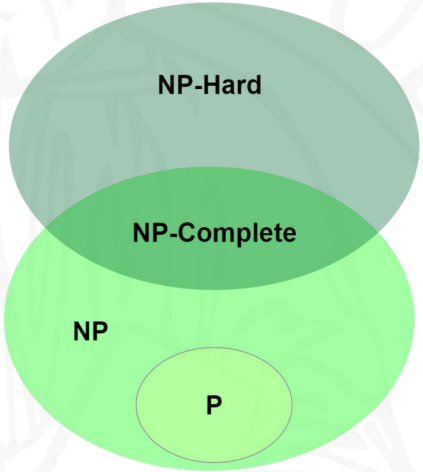
Complexity Classes: P, NP, and NP-Completeness

- **P (Polynomial):**
  Problems solvable in $O(n^k)$.

- **NP (Non-deterministic Polynomial):**
  Problems where a solution can be verified in **polynomial time**.

- **NP-Complete:**
  If a polynomial algorithm is found for one, then P=NP.

- **The Complexity Gap:** Moving from 2-SAT (in P) to **3-SAT** (NP-Complete) fundamentally changes the nature of the problem.

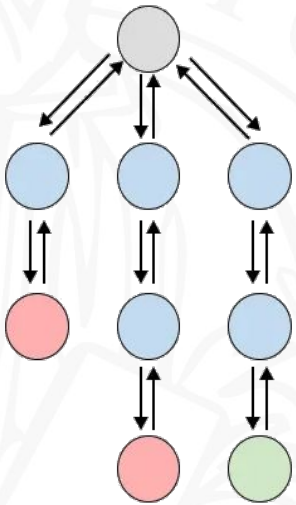


$$(p \vee \neg q \vee r) \wedge (q \vee r \vee \neg p) \wedge (\neg r \vee \neg p \vee q)$$

Backtracking:

A recursive procedure to find an assignment in CNF formulas:

1. **Base Case:** If the formula is empty → SAT.
   If it contains an empty clause → UNSAT.
2. **Branching:** Choose an unassigned variable x and set x = True.
3. **Simplification:**
   - Remove clauses where x appears (satisfied).
   - Remove ¬x from remaining clauses (false literal).
4. **Recursion:** If it fails, backtrack and try x=False.



Optimization: Unit Propagation Heuristic

Concept:
If a clause contains only one unassigned variable, its value is forced.

Effects: $(\neg q) \wedge (q \vee r) \wedge (\neg r \vee \neg q) \longrightarrow (r)$

- Remove all clauses containing the literal in its positive form.
- Remove the literal from all clauses where it appears as a negation.

➔ Drastically prune the tree.

The M/N Parameter

We can define the "density" of the system using the ratio:

$$\alpha = \frac{\#Clauses = M}{\#Variables = N}$$

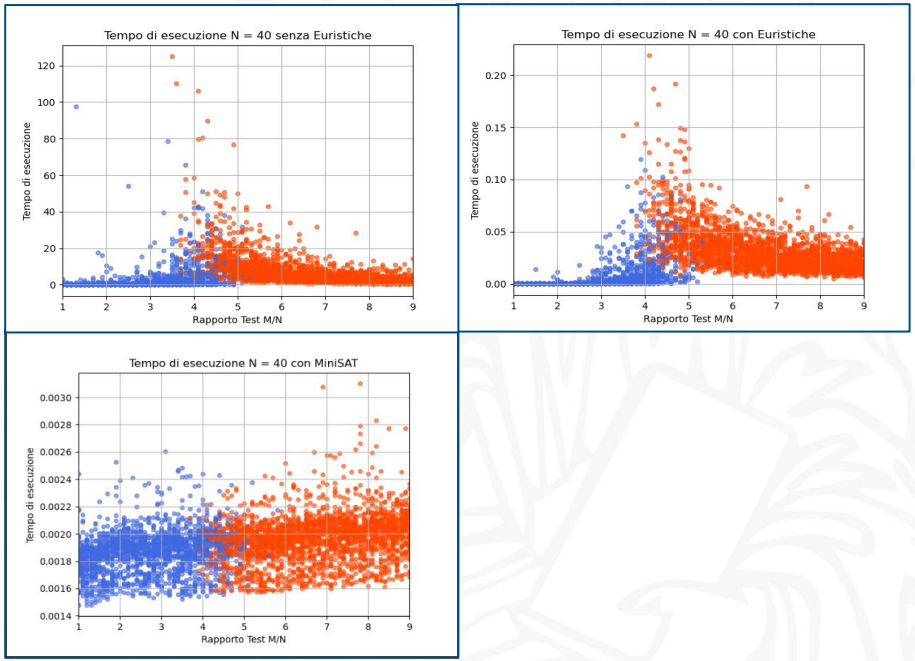
- **Under-constrained (Low α):**
  Many solutions exist, the problem is "easy" SAT.
- **Over-constrained (High α):**
  Too many constraints, the formula is likely "impossible" UNSAT.
- **Critical Region:** Where the sudden phase transition occurs.

Experimental Setup

- **Variables:** N={10,20,30,40}.
- **Range:** α varying from 1 to 9.
- **Methodology:** Generating random 3-SAT instances and solving them via:
  1. Backtracking.
  2. Backtracking + Unit Propagation.
  3. MiniSat (industrial solver).

Results: Average Execution Time

- **Observation:**
  Execution time is not monotonic.
- **The Difficulty Peak:**
  For α<4.2, times are low.
  For α>4.2, times decrease but remain higher than the easy regime.
- **Hardest Instances:**
  The most expensive formulas are found at the transition threshold.

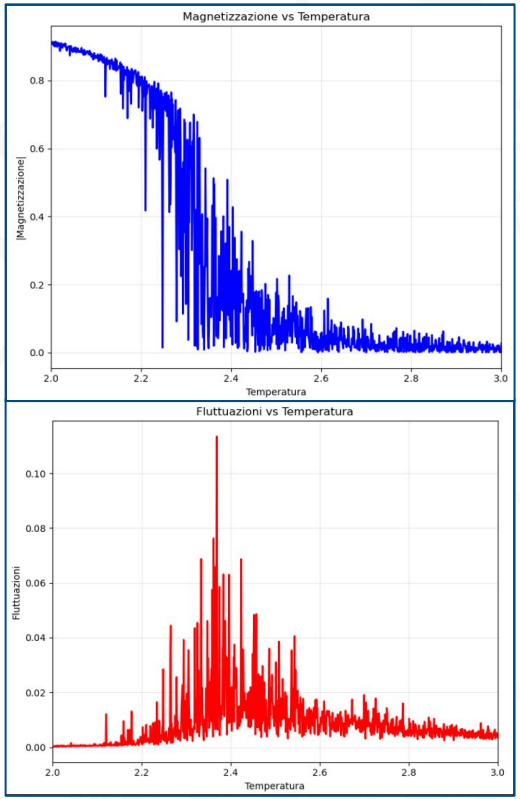


Physical Analogy: Phase Transitions

- **SAT Threshold:** Behaves like magnetization in the **Ising Model** at the Curie point.

- At α≈4.26, the probability of satisfiability drops from 1 to 0.

- **Thermodynamic Limit:** As N increases the transition becomes sharper (step function), typical of phase transitions.

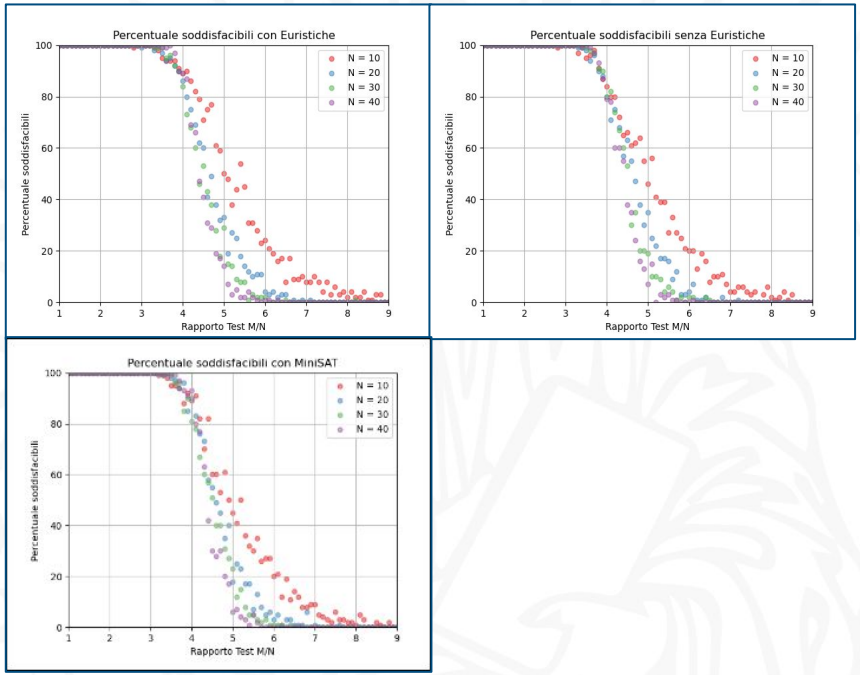


Satisfiability Percentage:

- **Visualization:**
  P(SAT) curves for different N intersect near α≈4.2.

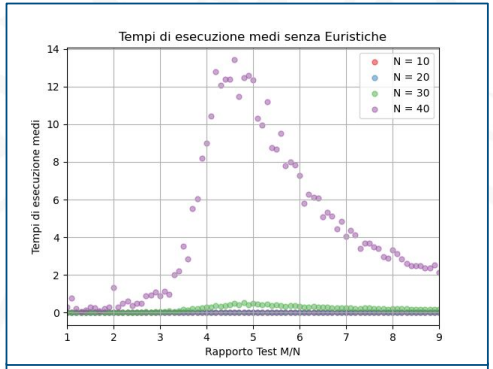
- **Interpretation:**
  For small N the transition is smooth, for large N it is sharp.

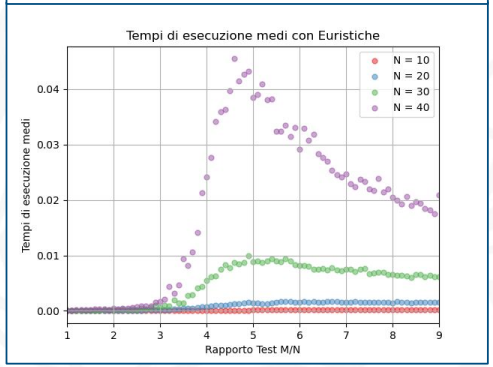
- The independence from the specific algorithm confirms that the transition is an **intrinsic property of the problem**.



Average Computational Cost and Steps:

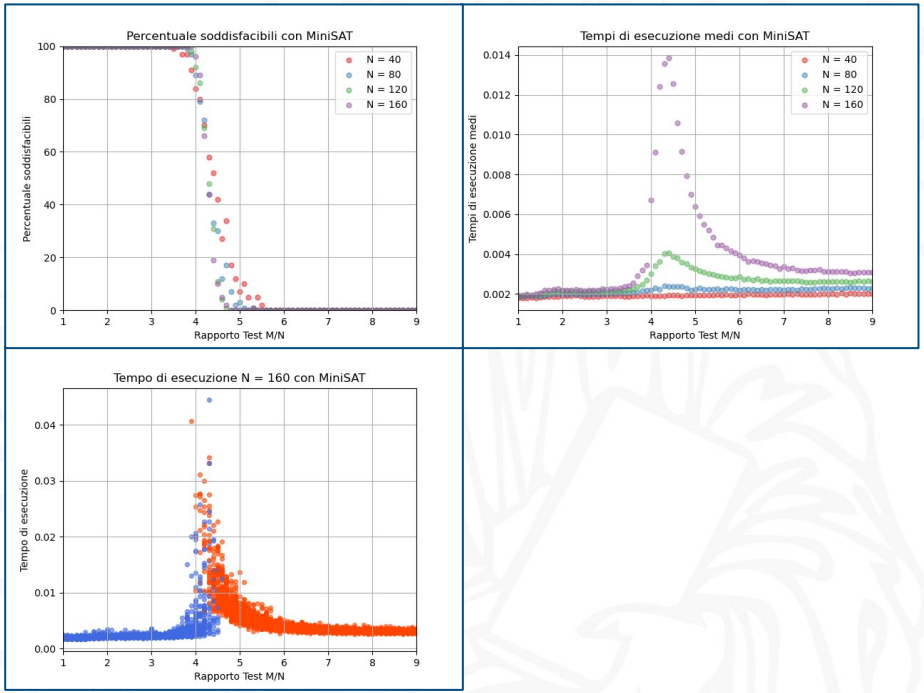
- The cost is minimal both in underconstrained and overconstrained regions.
- A peak appears near the critical region (M/N≈4.2)
- Heuristics do not move the peak's location, but they significantly reduce its magnitude.





Average Computational Cost and Steps:

- Minisat tourned out to be the best-performing method.
- We compared its performance by increasing both the number of variables.
- The results of the paper come out for a very large number of variables.



Conclusions:

- 3-SAT exhibits a phase transition governed by the ratio $\alpha = M/N$.
- The hardest instances are not those with the most constraints, but those at the "edge of chaos" ($\alpha \approx 4.2$).
- Strategies like Unit Propagation are essential to mitigate cost explosion.
- MiniSat is the most efficient due to advanced optimizations, yet it remains bound by the fundamental complexity of the problem.

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References:

- Franco Bagnoli - Lecture slides for statistical physics and complex systems.
- Niklas Eéen and Niklas Sörensson - Minisat: A sat solver with conflict-clause minimization.
- Brian Hayes - Can’t get no satisfaction.